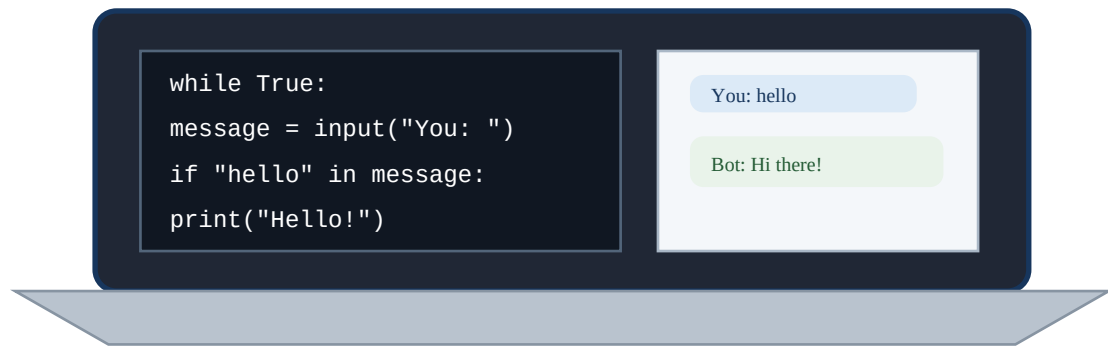


MINI ASSESSMENT

Build Your Own Rule-Based Python Chatbot

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Python chatbot development workflow

Real-world visual: Python code and a chatbot conversation running together.

1. Aim

To develop a simple rule-based conversational chatbot in Python that continuously accepts user messages, identifies predefined inputs, provides suitable responses, and exits when the user enters **bye** or **exit**.

2. Requirements / Concepts Used

The program uses **input()**, **.lower().strip()**, **while True**, **if / elif / else**, the **in** operator, and **break**. It asks for the user's name and uses the name in a response.

3. Algorithm

1. Start the program.
2. Ask the user for their name.
3. Display a welcome message.
4. Start an infinite loop using **while True**.
5. Read and clean the user's message using **lower().strip()**.
6. Check for exit commands.
7. Check greetings and predefined questions/topics.
8. Give an appropriate response.
9. Use **else** for unknown messages.
10. Repeat until **bye** or **exit** is entered.

4. Python Program

```
name = input("Enter your name: ").strip()

print(f"Hello {name}! I am your Python chatbot.")
print("You can ask me about Python, study, or the weather.")
print("Type 'bye' or 'exit' to end the chat.")

while True:
```

```
message = input("You: ").lower().strip()

if message in ("bye", "exit"):
    print(f"Goodbye {name}! Have a great day.")
    break

elif "hello" in message or "hi" in message or "hey" in message:
    print(f"Hello {name}! Nice to chat with you.")

elif "python" in message:
    print("Python is a beginner-friendly programming language.")

elif "study" in message or "exam" in message:
    print("Make a study plan, practice regularly, and revise important topics.")

elif "weather" in message:
    print("I cannot check live weather, but you can use a weather app.")

elif "who are you" in message or "your name" in message:
    print("I am a simple rule-based Python chatbot.")

else:
    print("Sorry, I don't understand that yet. Try Python, study, or weather.")
```

5. Explanation of the Program

Name Input: The program first asks for the user's name and stores it in **name**.

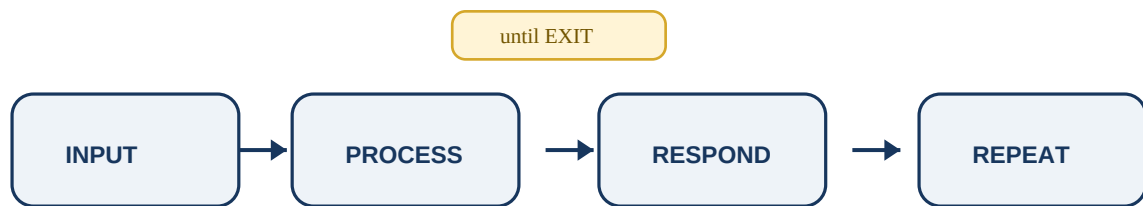
Continuous Conversation: The **while True** loop keeps accepting messages until an exit command is entered.

Input Processing: **lower().strip()** makes inputs such as **HELLO** easier to recognize.

Decision Making: **if / elif / else** checks greetings, Python, study/exam, weather, identity questions, and unknown messages.

Flexible Matching: The **in** operator allows phrases such as **hello bot** or **hey hello** to trigger a greeting.

Exit: **bye** or **exit** prints a goodbye message and **break** stops the loop.



Target flow: INPUT → PROCESS → RESPOND → REPEAT → EXIT

6. Sample Test Cases

Input	Expected Response
hello / HELLO	Greeting response
hi	Greeting response
hello bot	Greeting response
python	Python information
study / exam	Study advice
weather	Weather-related response
random message	Unknown-message response
bye / exit	Goodbye message and program ends

7. Result

The rule-based Python chatbot was successfully developed. It accepts continuous input, recognizes multiple greetings and topics, handles unknown messages, uses the user's name, and exits correctly using **break**.

8. Conclusion

This mini project demonstrates the basic working principle of a rule-based conversational chatbot using Python loops, input processing, and conditional statements. The chatbot can be extended with more rules and topics.